

Dravid K

Python Developer | Generative AI & LLM Applications for EdTech | Chennai, India

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Summary

Python developer with 1.5+ years of experience building production Generative AI applications for digital learning at TNQ Technologies: multi-stage LLM pipelines (Claude, OpenAI) that generate lesson plans, question banks and content for Tamil Nadu's Class 6–12 curriculum, plus FastAPI services, NLP preprocessing and speech ML (WhisperX, wav2vec2).

Technical Skills

GenAI & NLP: LLM APIs (Anthropic Claude, OpenAI), Prompt Engineering, Multi-step LLM Pipelines, Structured JSON Outputs, Output Validation & Evaluation, Prompt Caching, Streaming, Tamil Text Processing
ML & Speech: WhisperX (ASR, forced alignment), wav2vec2 (Hugging Face / PyTorch), Text-to-Speech
Backend & APIs: Python, FastAPI (async), Pydantic, Django, Node.js, Express.js, REST APIs, JWT/OAuth
Data: ETL, EPUB/PDF Extraction (BeautifulSoup, pdfplumber), SQL (PostgreSQL, MySQL, MariaDB)
DevOps & Tools: Git, GitHub, Linux, Nginx, Apache, AWS EC2, Linode, Grafana | JavaScript, HTML, CSS

Professional Experience

Python Developer

TNQ Technologies

May 2025 – Present

Chennai, Tamil Nadu

SK-CAP — GenAI Content Automation Platform — *Python, FastAPI, Claude API, BeautifulSoup, Node.js*

- **GenAI platform:** Built the Python/FastAPI + Claude API platform generating content pages, 100-question banks and multi-day lesson plans for Classes 6–12 (Social Science, Maths, English, Science, Biology, Physics) on Tutorea.
- **Multi-stage LLM pipelines:** Designed section-extraction → day-allocation → per-day generation chains with JSON outputs validated and auto-repaired in Python; turned a 144K-character Physics chapter into a 22-day, teacher-ready lesson plan with zero failed calls.
- **Reliability & cost:** Added streaming, mid-stream retry with backoff, prompt caching and per-section context slicing (~70% fewer input tokens per call); validation and top-up logic guarantee 100 questions per chapter (was 96–98).
- **Prompt engineering & evaluation:** Encoded teacher-approved pedagogy (concept checks, Tamil scaffolding, numerical routines, MCQ-to-5-mark formats) into reusable prompt modules; audit scripts check structure, formulas and answer quality.
- **Data preprocessing:** Replaced PDF extraction (pdfplumber, Document AI), which failed on two-column layouts, with EPUB parsing on navigation-anchor boundaries; cleanup preserves physics notation and trims back matter.
- **Architecture & collaboration:** Data-driven subject routers and builder registry scaled to 7 subjects, including split-volume textbooks, with regression tests before each release; wrote an external UX agency's integration spec.

Tamil Read-Along Pipeline (Speech ML) — *Python, WhisperX, wav2vec2, PyTorch, Linode*

- Built a pipeline that aligns narration audio to Tamil text at word level using WhisperX with a Tamil wav2vec2 model, producing synchronized karaoke-style reading pages; ran it on two Linode VMs, backed by a 47-check regression suite, and delivered a 70-file client batch on deadline.

Textbook Platform, APIs & Learning Features — *FastAPI, Node.js, Express.js, MariaDB, JavaScript*

- Architected a config-driven system using JSON manifests to serve 765+ textbooks and 10,000+ pages; built an async FastAPI extraction microservice serving 20+ concurrent users; caching cut redundant downloads by 80%.
- Built a Node.js/MariaDB exercise-submission backend with debounced autosave and row-level locking, fixed a JWT role bug across email/password and Google OAuth login, and added W3C Web Annotation across 10,000+ pages.

Python Full-Stack Developer Intern

Accent Techno Soft

Dec 2024 – Apr 2025

Coimbatore, Tamil Nadu

- Built a sports event management system (Django, PostgreSQL) with role-based access control; optimized ORM queries (`select_related`, `prefetch_related`) and indexing to cut dashboard query time by 40%.

Projects & Writing

- **AI-Powered Education Query Platform** (Django, OpenAI API): LLM-based assistant answering student questions on the Samacheer Kalvi syllabus by class, subject and query type, with rate limiting and secure API-key handling.
- **Technical writing** (Medium, 2026): Flask on a VPS with Apache; Grafana behind an Apache reverse proxy.

Education & Achievements

Master of Computer Applications (MCA) — Kongu Arts & Science College, CGPA 8.2/10

2023 – 2025

B.Sc. Computer Science — Kongu Arts & Science College, CGPA 7.57/10

2020 – 2023

- **Cricket:** Captained Tiruppur District U14/U16/U19 teams; Tamil Nadu Colts (2023); 5+ years in TNCA league.